

Freight Rate Prediction

ML Engineering Technical Assessment | Tania Iveth Giron Espana

1. Problem Statement

Predict posted_rate (USD) for 12,000 freight loads in validation.csv using 48,000 historical loads as training data. A secondary deliverable requires forecasting daily rates for a fixed route (Lexington to Fort Wayne, 360 mi, Dry Van, 32,000 lb) across all 31 days of December 2025.

2. Data Overview

Dataset	Rows	Date range	Target available
train-test.csv	48,000	Jan - Oct 2025	Yes
validation.csv	12,000	Nov 2025+	No (predict)
december-chart-inputs.csv	31	Dec 2025	No (predict)

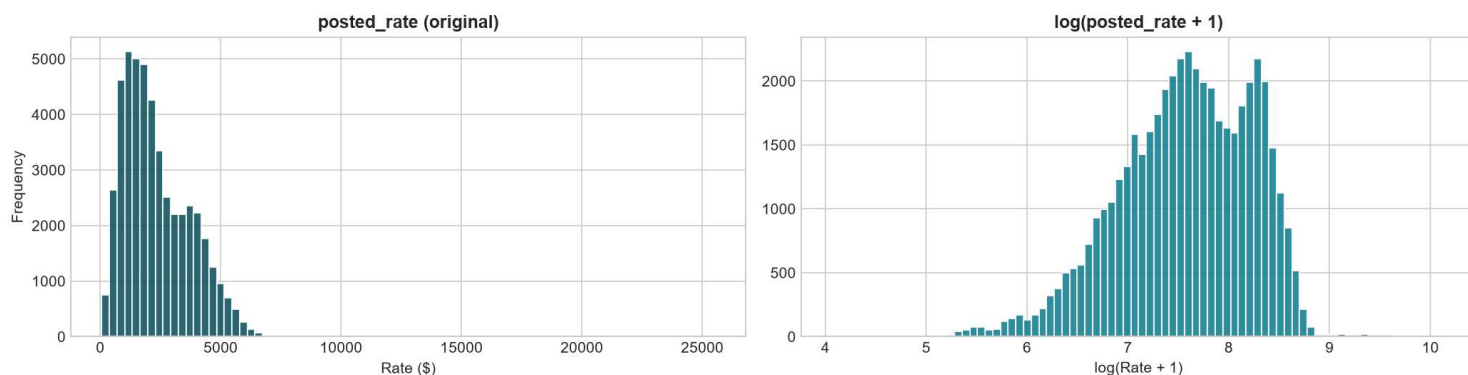
Missing values: weight (300 rows, 0.6%) and market_index (374 rows, 0.8%). Imputed with training-set medians to prevent data leakage into the model.

3. Exploratory Data Analysis

3.1 Target Distribution

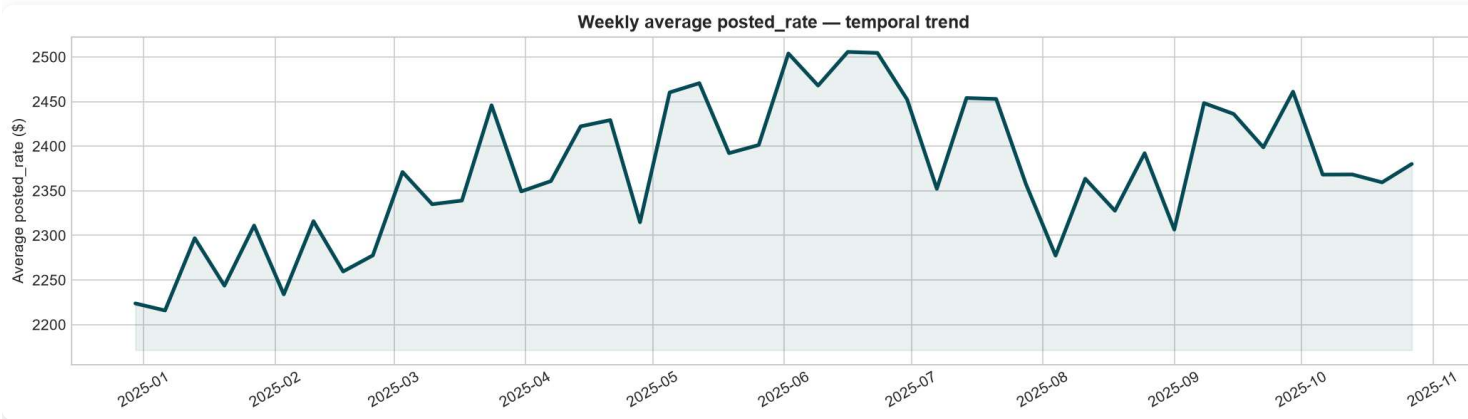
The target is right-skewed (skewness ~2.0). A log transform reduces skewness to ~0.2 and stabilizes variance. The model is trained on $\log(\text{posted_rate} + 1)$ and predictions are reversed with $\exp m1()$ at inference time.

Log-transform target?



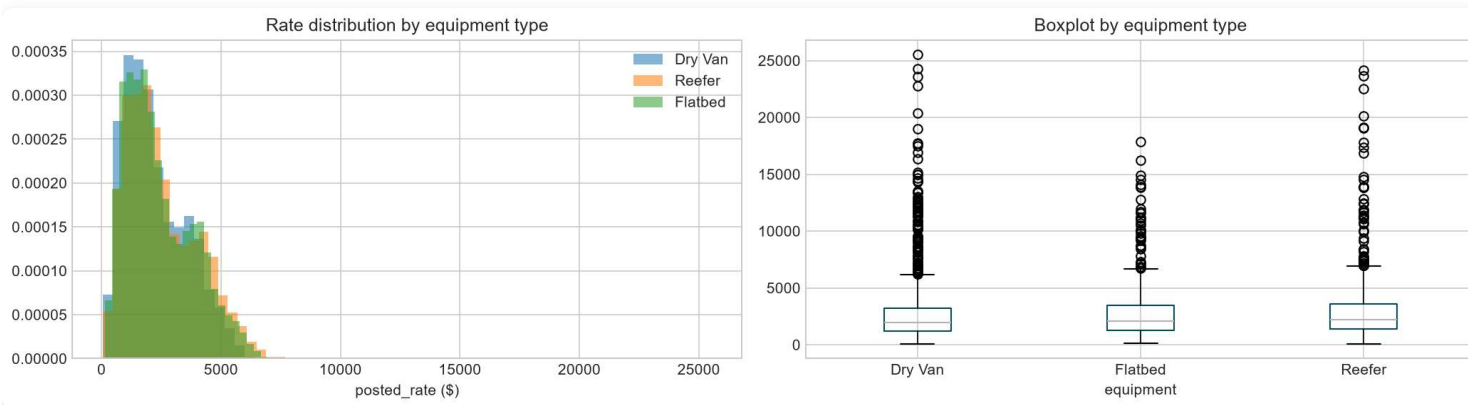
3.2 Temporal Trend

Weekly averages show clear seasonality: peaking in June (~\$2,500), dipping in August, and recovering in Q4. This confirms the necessity of a time-based split.



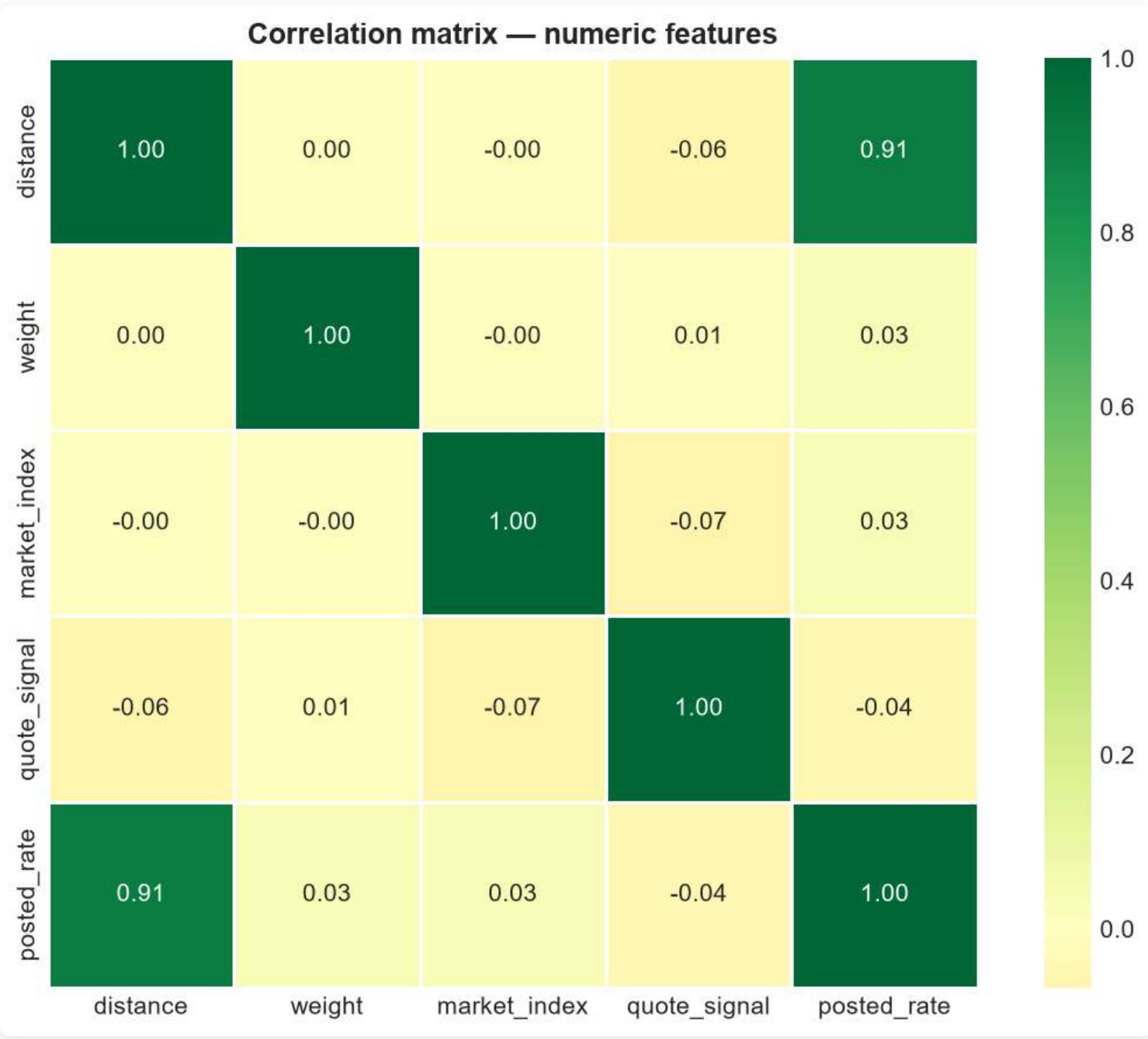
3.3 Equipment Type

Rate differences between equipment types are modest (Reefer: \$2,554 avg vs Dry Van: \$2,272), suggesting equipment alone is not the primary pricing driver.



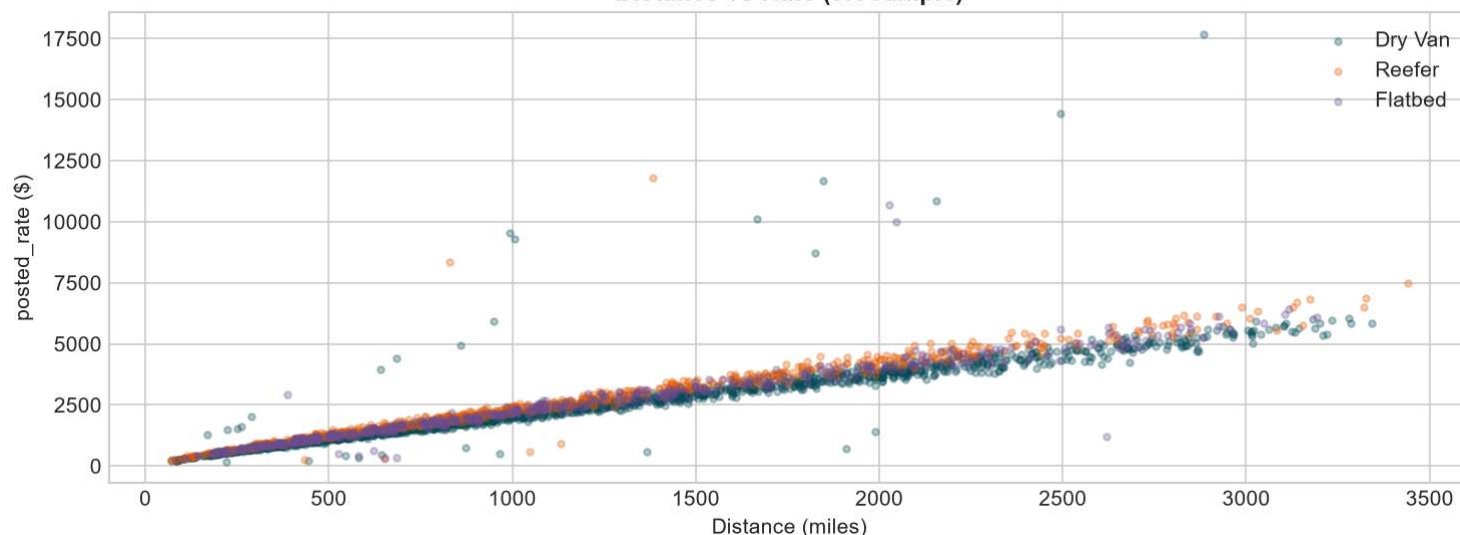
3.4 Correlation Matrix

Distance dominates with 0.91 correlation with posted_rate. market_index and quote_signal show near-zero linear correlation - their signal is likely non-linear, which tree-based models capture effectively.



3.5 Distance vs Rate

Distance vs Rate (3K sample)



4. Data Split Strategy

Temporal split - not random. A random split would allow training on October data while evaluating on August data, leaking future information and inflating metrics. The temporal split simulates the real production scenario: train on the past, predict the future.

Set	Period	Rows	Purpose
Training	Jan - Sep 2025	43,147	Model learning
Hold-out	Oct 2025	4,853	Error estimation / hyperparameter tuning
Final validation	Nov 2025+	12,000	Submission - never used for decisions

5. Feature Engineering

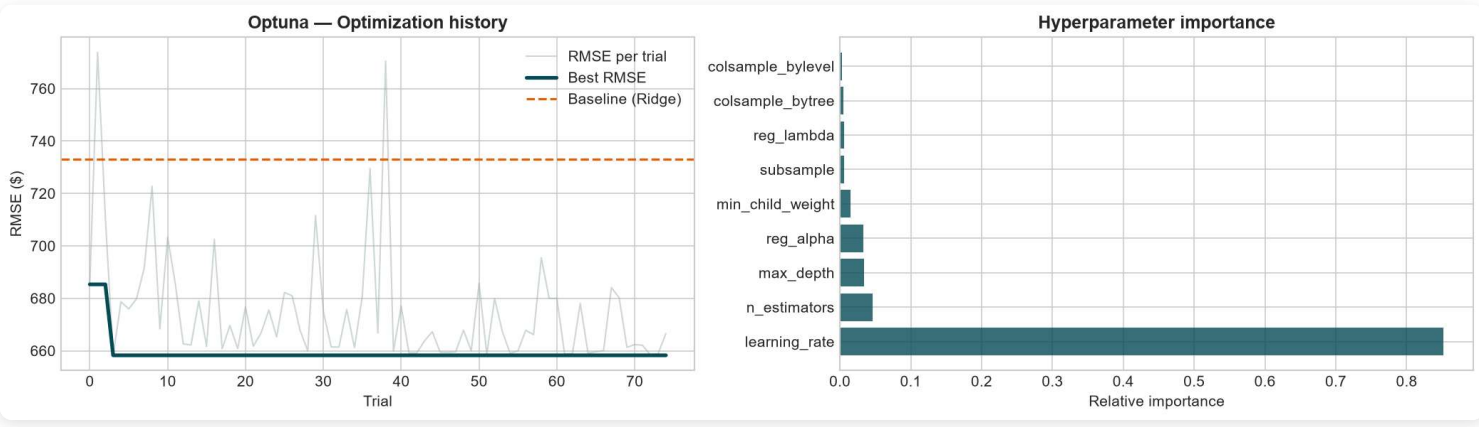
Category	Features	Rationale
Distance	distance, distance ² , haversine_dist, route_sinusosity	Non-linear distance effects; haversine captures route direction
Route stats	route_rate_mean, route_rpm, route_rpm_p75, route_rpm_std	Historical \$/mile by (pickup, delivery) pair

City encoding	pickup_rate_mean, delivery_rate_mean	Target encoding computed on training set only
Market signals	market_index, quote_signal, market_x_quote	Interaction captures joint market effect
Temporal	day_of_week, month, week_of_year, is_weekend, quarter	Seasonal freight demand patterns
Geography	lat/lon coordinates, 3x3 region grid	East-west lane differentiation

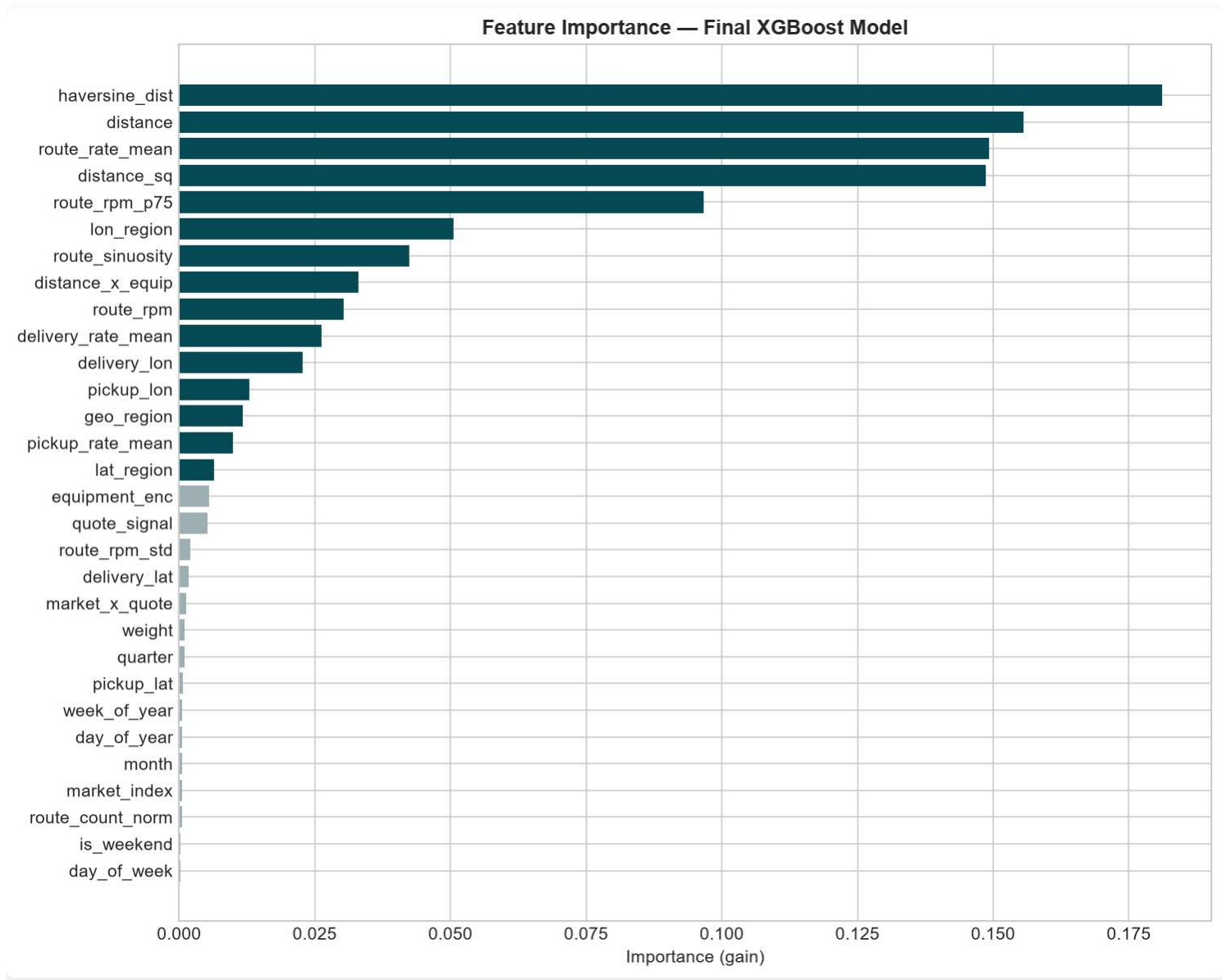
Key finding: haversine_dist (straight-line geographic distance) emerged as the #1 most important feature. It implicitly encodes route direction, capturing lane-level pricing pressure that road distance alone does not reflect.

6. Model: XGBoost + Optuna

Optuna ran 75 trials with Bayesian optimization (TPE sampler) to minimize hold-out RMSE. The model trains on log-transformed targets with squared error loss. After tuning, the final model was retrained on the full 48,000-row dataset.



Feature Importance

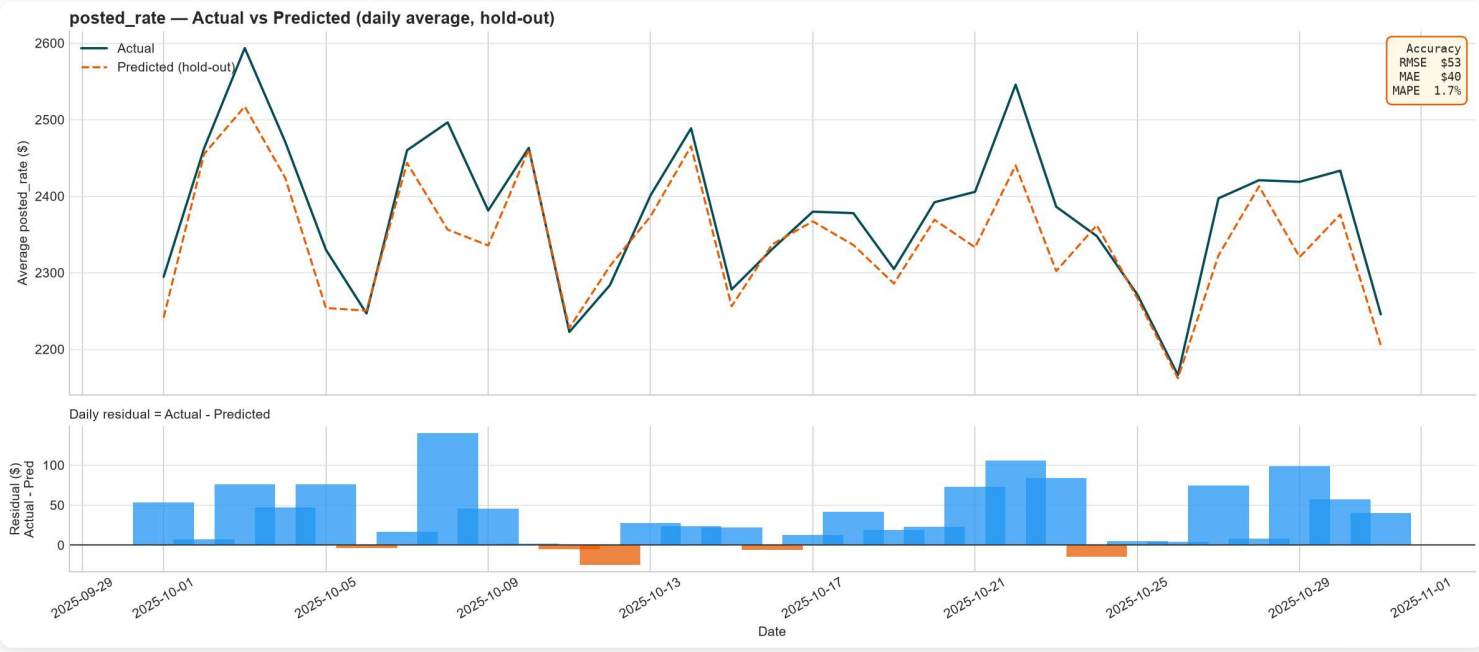


7. Validation Results

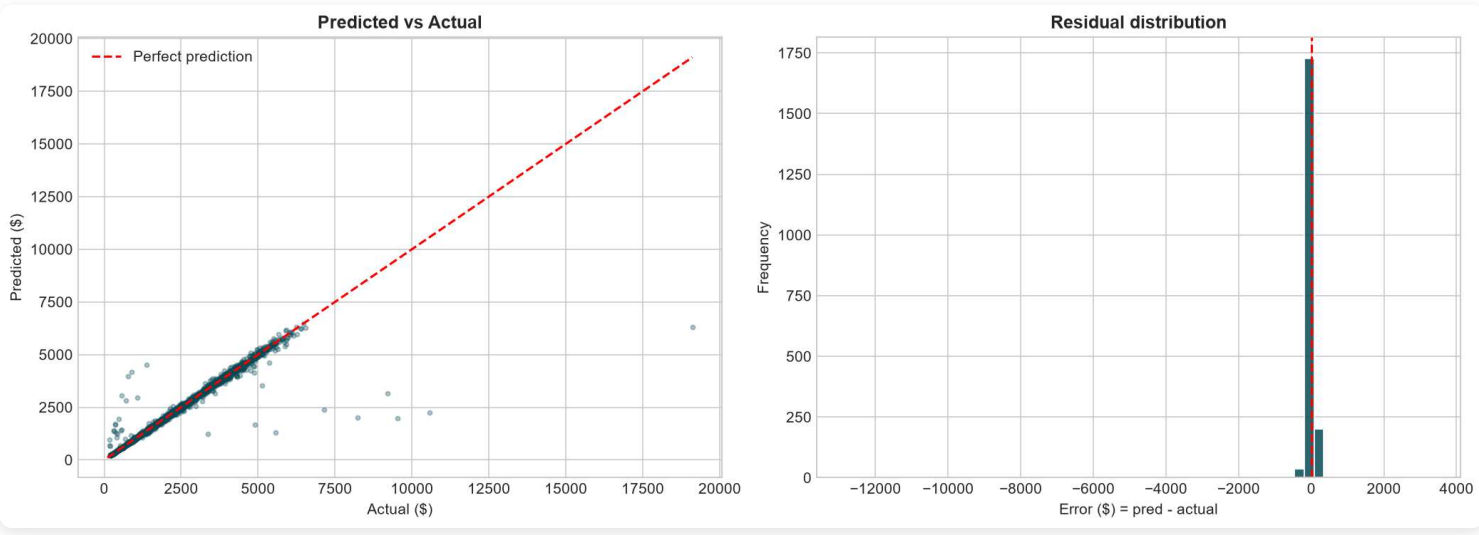
	\$640 RMSE	\$112 MAE	\$32 Median AE	4.7% WMAPE	0.83 R2
Metric	Baseline (Ridge)		XGBoost + Optuna		
RMSE (\$)	\$733		\$640		
MAE (\$)	\$221		\$112		

Median AE (\$)	-	\$32
WMAPE (%)	9.3%	4.7%
R2	0.77	0.83

Actual vs Predicted - Daily Average (Hold-out: October 2025)



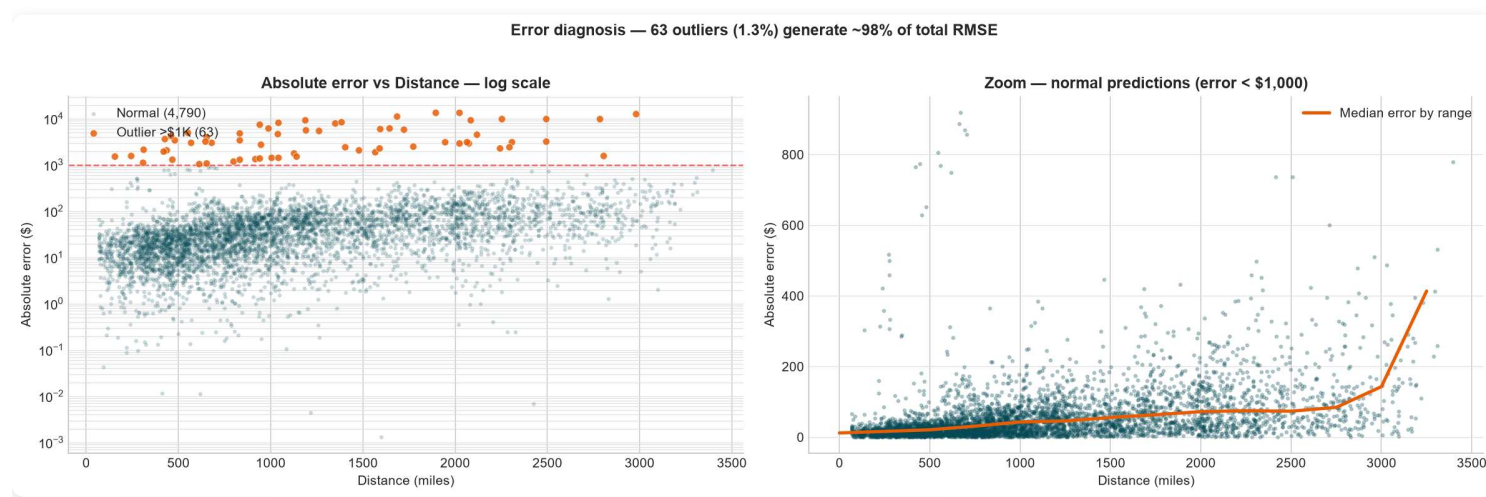
Residual Distribution



8. Outlier Analysis

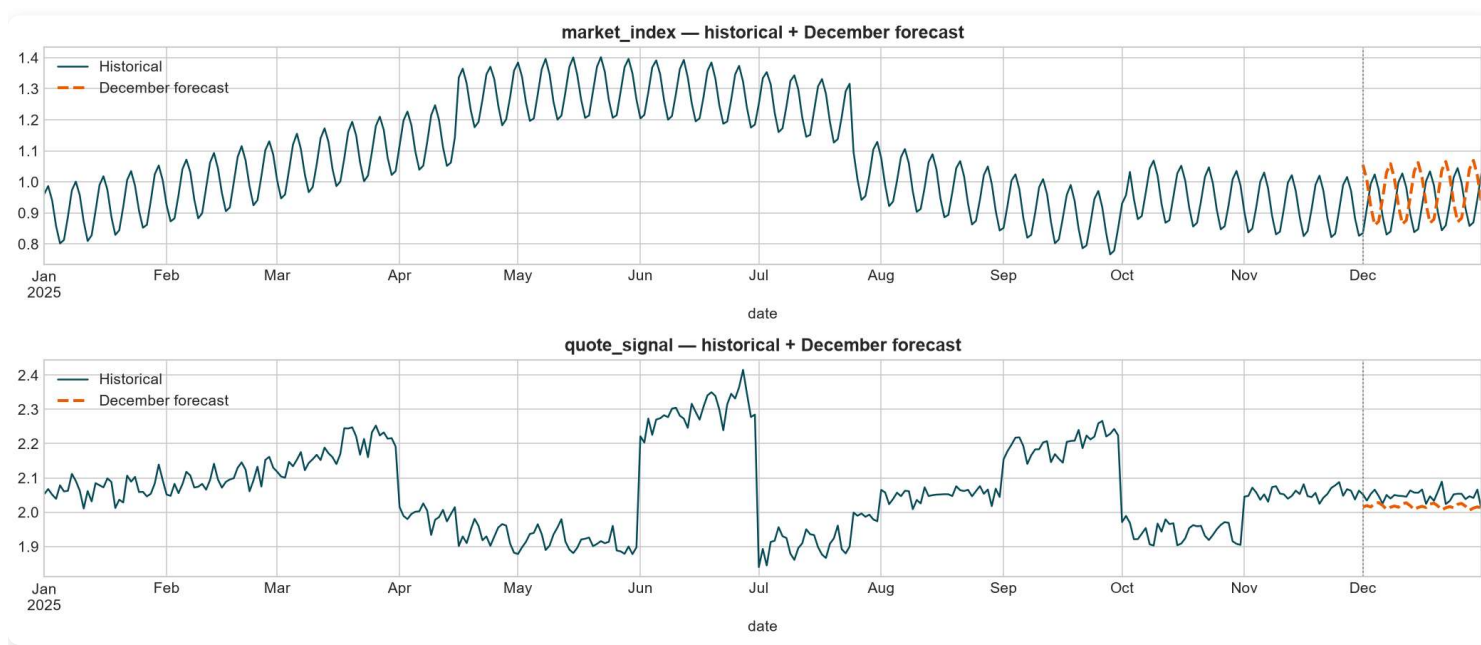
63 loads (1.3%) drive 98% of total squared error. The remaining 98.7% of predictions have a median error of just \$32.

Outlier loads have rate-per-mile of \$5-\$12 vs the typical \$2/mile. They appear across all distance ranges and equipment types, consistent with spot market surges or backhaul imbalances not captured by the available features. The model is not systematically biased - it simply cannot predict exceptional market events without additional contextual signals.

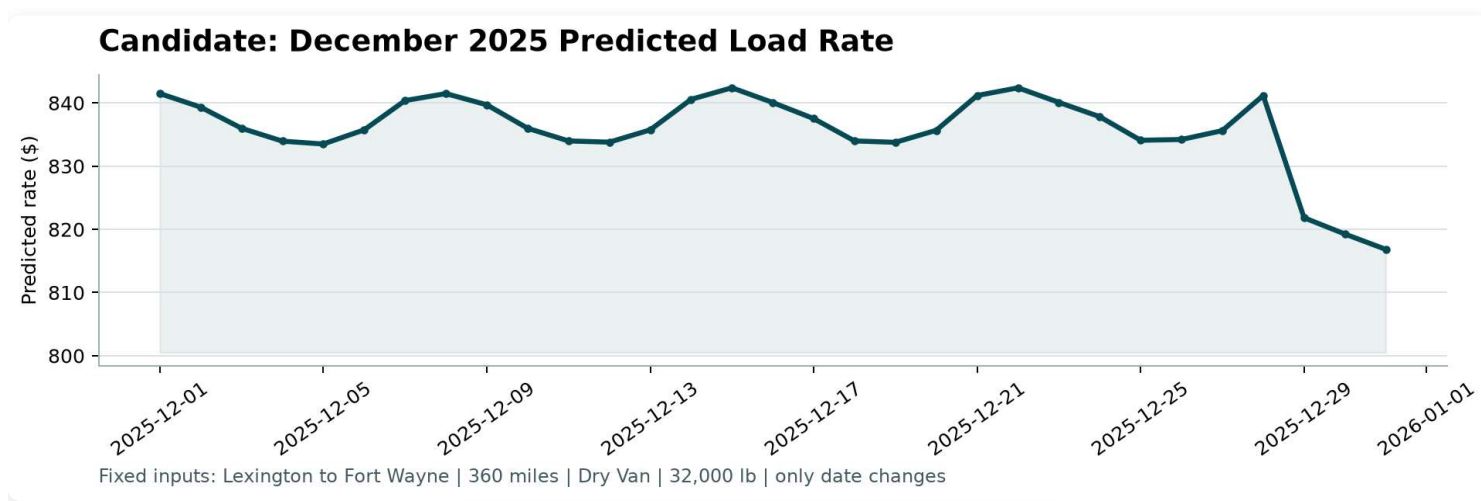


9. December 2025 Forecast - Lexington to Fort Wayne

The december file lacks `market_index` and `quote_signal`. Exponential Smoothing with additive trend and weekly seasonality (period=7) was applied to the full Jan-Nov 2025 historical series to generate day-specific forecasts for December, preserving the weekly cycle rather than imputing a flat average.



Predicted rates range from approximately \$817-\$842, consistent with the route's historical rate-per-mile (~\$2.3/mile) and the weekly demand cycle.



10. Limitations and Future Work

- **Outlier loads:** 63 loads with anomalous \$/mile are unpredictable with current features. Potential solutions: lane-level demand signals, fuel surcharge data, or a two-stage model that separately prices premium loads.
- **market_index signal:** Near-zero linear correlation and minimal feature importance suggest its effect may be lagged or indirect.
- **Long-haul precision:** Prediction error grows beyond 2,500 miles due to data sparsity on transcontinental routes.

- **December uncertainty:** One-month-ahead market index extrapolation introduces forecast uncertainty. Quantile regression would better communicate prediction intervals.